A TECHNICAL REPORT ON

STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES)

CARRIED OUT AT

ELROL INTEGRATED SERVICES LTD

17 MURTALA MUHAMMED AIRPORT ROAD, AJAO ESTATE, LAGOS STATE.

PERIOD OF ATTACHMENT:

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BY

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SUBMITTED TO:

DEPARTMENT OF STATISTICS

FACULTY OF PHYSICAL SCIENCE

NNAMDI AZIKIWE UNIVERSITY, AWKA.

IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE, STATISTICS.

DEDICATION

This report is dedicated to God Almighty, who gave me protection, guidance, grace, and wisdom all the way through in carrying out my Industrial Training (IT) exercise, and to my lovely family and good friends.

ACKNOWLEDGEMENTS

I begin by thanking Almighty God for His unending grace, and wisdom throughout my SIWES program. Every step of this journey was guided by His hand from the start to the very end and I strength remain deeply grateful for the peace and clarity He gave me along the way.

My sincere appreciation goes to the management of Elrol Integrated Services Ltd for accepting me as an intern and allowing me to learn, explore, and contribute meaningfully to the company. I'm especially grateful to my supervisor, Mr. Emmanuel Chukwuma, for his patience, guidance, and mentorship that shaped my understanding of real business operations and analytics. The experience I gained while developing sales records, inventory systems, analyzing sales and records data, purchase order tracking tools, Power BI dashboards, and even an AI chatbot support system will always remain a strong foundation in my professional growth.

I also extend my gratitude to the Industrial Training Fund (ITF) and Nnamdi Azikiwe University for organizing and supporting this program, which bridges the gap between theory and practice.

To my parents and family, your love, prayers, and unwavering belief in me kept me going even when things got tough. You've been my constant source of strength, encouragement, and inspiration and I dedicate this milestone to you.

Finally, I appreciate everyone who, in one way or another, contributed to my learning experience colleagues, friends, and mentors who shared knowledge, advice, or support. This journey has been more than training; it's been a period of growth, discipline, and discovery that I will always value.

ABSTRACT

This Students' Industrial Work Experience Scheme (SIWES) report documents the practical training I undertook at Elrol Integrated Services Ltd as part of the requirements for the award of a B.Sc. in Statistics. The training exposed me to real business operations, especially in the areas of sales analysis, inventory management, procurement tracking, and business reporting.

During the programme, I applied several statistical and analytical techniques, such as data collection and organization, data cleaning, descriptive analysis, trend analysis, correlation, and dashboard development using tools like Microsoft Excel and Power BI. I also participated in building operational systems, including a Sales Record, a Purchase Order (PO) Tracking System, an Inventory Record, and a live Power BI dashboard connected to Excel/OneDrive for real-time monitoring. These experiences strengthened my understanding of statistical principles and improved my analytical, technical, and problem-solving skills.

This report highlights the tasks performed, the statistical methods applied, the challenges encountered, and the professional competencies gained. The SIWES programme effectively bridged the gap between classroom knowledge and real-world applications, contributing significantly to my development as a data-driven problem solver.

TABLE OF CONTENTS

TABLE OF FIGURES 5
CHAPTER ONE: 6
SIWES 6
1.1 Background of the Study 6
1.2 Brief History of SIWES 6
1.3 Objectives of the SIWES Programme 6
1.4 Significance of SIWES 7
1.5 Importance of Industrial Training to Statistics Students 7
CHAPTER TWO: 8
ORGANIZATIONAL PROFILE 8
2.1 Overview of Elrol Integrated Services Ltd 8
2.2 Organizational Structure of the Establishment 8
2.3 Description of the Department Attached To 9
CHAPTER THREE: 10
METHODOLOGY 10
3.1 Introduction 10
3.2 Aim Objectives 10
3.3 Data Collection 10
3.4 Methodology 12
3.4.1 Descriptive statistics: 12
3.4.2 Trend Analysis: 12
3.4.3 Pareto Analysis 13
3.5 Systems Developed 15
3.6 Tools and Software Used 16
CHAPTER FOUR: 18
DATA ANALYSIS 18
4.1 Introduction 18
4.2 Data Preparation and Cleaning 18
4.5 Exploratory Data Analysis 18
4.6 Interpretation of Trend Analysis 19
4.7 Interpretation of Results 20
CHAPTER FIVE: 21
CONCLUSION & RECOMMENDATION 21
5.1 Summary of Experiences: 21
5.2 Skills Acquired: 21
5.3 Conclusion: 21
5.4 Recommendations: 22
REFERENCES 23

TABLE OF FIGURES

Figure 1: Hard copy of an invoice 11
Figure 2: Pareto Chart showing customer contribution to sales 13
Figure 3: Ranking of customers based on their sales contribution 14
Figure 4:Table of customer segmentation from Pareto Analysis 15
Figure 5:Top selling products 17
Figure 6: Top customers by Net and gross sales 18
Figure 7: Powe-BI dashboard of the analyses 19

CHAPTER ONE:

SIWES

1.1 Background of the Study

The Student Industrial Work Experience Scheme (SIWES) is a structured training programme designed to prepare students of tertiary institutions for the transition from academic learning to professional practice. Initiated by the Industrial Training Fund (ITF) in 1973, SIWES was developed in response to the growing concern that graduates lacked the practical skills required by employers in various sectors of the Nigerian economy (Okeke, 2018). The programme serves as a bridge between theoretical instruction and real-world application, offering students the opportunity to gain hands-on experience in their respective fields.

1.2 Brief History of SIWES

SIWES began as a pilot project involving only a few institutions and disciplines, primarily engineering and technology. Over time, it expanded to include students from universities, polytechnics, and colleges of education across various fields such as science, agriculture, business, and statistics (Adebayo, 2020). The programme was born out of concerns from employers that graduates lacked the hands-on skills necessary for industrial roles. Today, SIWES is a mandatory component of many academic programmes, coordinated by the ITF in collaboration with the National Universities Commission (NUC), National Board for Technical Education (NBTE), and other regulatory bodies.

1.3 Objectives of the SIWES Programme

The primary objectives of SIWES are as follows:

I. To expose students to industrial environments and practices relevant to their field of study.

II. To enable students to apply theoretical knowledge in practical settings.

III. To develop students' technical, analytical, and interpersonal skills.

IV. To foster a deeper understanding of industrial operations, standards, and expectations.

v. To enhance students' readiness for employment and professional growth.

1.4 Significance of SIWES

SIWES plays a critical role in the academic and professional development of students. Its significance includes:

i. Bridging the gap between academic instruction and industrial practice.
ii. Enhancing students' confidence and competence in real-world -settings.
iii. Providing insights into career opportunities and industry expectations.
iv. Encouraging innovation, problem-solving, and adaptability.
v. Strengthening the link between institutions and industries for national development.

1.5 Importance of Industrial Training to Statistics Students

For students of statistics, SIWES offers invaluable exposure to the practical dimensions of data science and analytics. Its importance includes:

i. Application of statistical theories to real-world problems.
ii. Use of statistical tools such as SPSS, R, Python, and Excel in data analysis.
iii. Understanding of how statistical insights inform decision-making in business, health, government, and research.
iv. Development of critical thinking, precision, and data-driven communication skills.
v. Preparation for careers in data science, research, policy analysis, and consultancy.

CHAPTER TWO:

ORGANIZATIONAL PROFILE

2.1 Overview of Elrol Integrated Services Ltd

Elrol Integrated Services Ltd is a reputable company engaged in multiple sectors of logistics and procurement. Founded by Mr. Emmanuel Chukwuma, with the mission to provide seamless supply chain solutions, the company has expanded to cover air and sea freight services, procurement and distribution of goods—notably as an authorized distributor of SKF bearings—and customs brokerage.

The company has positioned itself as a reliable service provider in Nigeria, offering efficient freight handling, procurement services, and related business solutions to industrial clients and commercial enterprises. Its operations are supported by dedicated departments that specialize in their respective functions, ensuring streamlined processes and prompt service delivery.0



2.2 Organizational Structure of the Establishment

The organizational structure at Elrol Integrated Services Ltd is hierarchical, designed to optimize operational efficiency and accountability. At the top is the Managing Director (MD), who oversees all departments and strategic decision-making. The company primarily operates through the following key departments:

1. **Procurement Department**
    I. Responsible for sourcing and distributing products, including SKF bearings.
    II. Manages purchase orders, inventory, and customer transactions.
2. **Freight Department**
    I. Handles air and sea freight operations.
    II. Ensures safe and timely delivery of goods.
3. **IT Department**
    I. Focused on business system development, automation, and data management.
    II. Supports procurement and freight operations with digital solutions such as ERP

systems, dashboards, and analytics tools.

4. **Accounting Department**

I. Manages financial records, payments, invoicing, and budgeting.

II. Tracks revenue, expenses, and other financial metrics for decision-making.

2.3 Description of the Department Attached To

During my Industrial Training, I was attached primarily to the Procurement Department, while also collaborating with the IT Department on system development. The Procurement Department's main functions include:

I. Receiving customer orders and processing purchase requests

II. Managing sales records and generating invoices

III. Monitoring stock availability and coordinating deliveries

IV. Liaising with suppliers for timely procurement of goods

Within this department, my role involved data analytics and system development, focusing on:

I. Analyzing sales records to identify top customers, high-value products, and sales trends

II. Creating automated Purchase Order (PO) tracking systems

III. Developing inventory management systems for real-time stock tracking

IV. Implementing a Power BI dashboard for management to visualize sales and operational metrics

CHAPTER THREE:

METHODOLOGY

3.1 Introduction

This chapter presents the methods, tools, and statistical techniques used for analyzing the operational data obtained during the Industrial Training at Elrol Integrated Services Ltd. The datasets used for the analysis were internal records from the company's Sales and Inventory in the Procurements department. These datasets were originally stored in Excel sheets and paper-based logs before being digitized, cleaned, and analyzed.

The purpose of the analysis was to understand sales performance, track procurement activities, manage inventory movement, interpret stock arrivals, and support management decision-making using descriptive statistics, trend analysis, correlation, and KPI computations. In addition, several digital systems were developed in Excel and Microsoft OneDrive to improve workflow efficiency, reduce manual work, and eliminate reporting errors.

3.2 Aim Objectives

Aim

The aim of the analysis is to evaluate business operations using statistical techniques and to build digital systems that enhance reporting accuracy, inventory control, procurement tracking, and overall workflow efficiency at Elrol Integrated Services Ltd.

Objectives

I. To clean, structure, and organize sales, procurement, and inventory datasets for easy analysis.

II. To perform descriptive and trend analysis on the company's operational data.

III. To identify relationships between key variables using correlation where necessary.

IV. To compute Key Performance Indicators (KPIs) that reflect business performance.

V. To develop digital systems for Sales Recording, Purchase Order Tracking, Inventory Management, and Stock Arrival Tracking.

VI. To build a Power BI dashboard for real-time visualization and decision-making.

VII. To streamline business operations and reduce reliance on manual records.

3.3 Data Collection

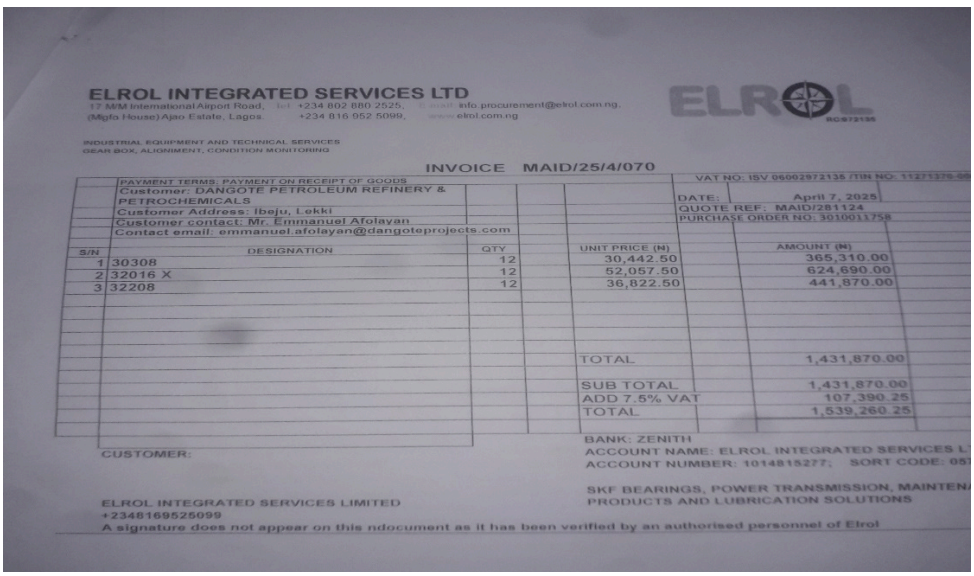
The data for this study were obtained as secondary data, collected from internal company records, including:

- Hardcopy invoices from customers
- Existing sales records in Excel maintained by the company
- Purchase orders (POs) and stock movement records
- Interviews with staff to understand procurement and sales processes

The collected data included the following fields:

- Invoice Date
- Supply Date
- Invoice Number
- Customer Name
- Product Description
- Quantity
- Unit Price
- Amount
- VAT Amount
- Withholding Tax
- Total Amount

The raw data was digitized and consolidated into a centralized Excel workbook to facilitate analysis.



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INDUSTRIAL EQUIPMENT AND TECHNICAL SERVICES
GEAR BOX, ALIGNMENT, CONDITION MONITORING

INVOICE MAID/25/4/070

PAYMENT TERMS: PAYMENT ON RECEIPT OF GOODS
Customer: DANGOTE PETROLEUM REFINERY & PETROCHEMICALS
Customer Address: Ibeju, Lekki
Customer contact: Mr. Emmanuel Afolayan
Contact email: emmanuel.afolayan@dangoteprojects.com

VAT NO: ISV 06002972135 /TIN NO: 11271379-0
DATE: April 7, 2025
QUOTE REF: MAID/281124
PURCHASE ORDER NO: 3010013758

S/N	DESIGNATION	QTY	UNIT PRICE (N)	AMOUNT (N)
1	30308	12	30,442.50	365,310.00
2	32016 X	12	52,057.50	624,690.00
3	32208	12	36,822.50	441,870.00
			TOTAL	1,431,870.00
			SUB TOTAL	1,431,870.00
			ADD 7.5% VAT	107,390.25
			TOTAL	1,539,260.25

CUSTOMER:

BANK: ZENITH
ACCOUNT NAME: ELROL INTEGRATED SERVICES L
ACCOUNT NUMBER: 1014815277; SORT CODE: 05

SKF BEARINGS, POWER TRANSMISSION, MAINTEN
PRODUCTS AND LUBRICATION SOLUTIONS

ELROL INTEGRATED SERVICES LIMITED
+2348169525099
A signature does not appear on this ndocument as it has been verified by an authorised personnel of Elrol

Figure 1: Hard copy of an invoice

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3.4 Methodology

During the training period, the statistical technique used for analysis includes the following:

3.4.1 Descriptive statistics:

Descriptive Statistics is a branch of statistics that involves summarizing, organizing, and presenting data in a meaningful way. Its primary purpose is to describe the main features of a dataset, providing a concise and comprehensible overview.

3.4.2 Trend Analysis:

Trend analysis is a statistical technique used to examine patterns or tendencies in data over a period of time. In this report, it was applied to the company's sales data to visualize how sales performance changed across months. Line charts were used to illustrate these trends, allowing insights into peak sales periods, fluctuations, and seasonal variations.
This method provides a quantitative view of historical performance and supports data-driven decision making, such as identifying periods for marketing focus or inventory planning.

Trend analysis involves studying patterns in data over time to detect increases, decreases, seasonality, and cyclical behavior.

This technique was used to understand how sales, procurement, and inventory change over months.

During the analysis, Trend analysis was used for the following purposes:

- Monitoring monthly sales patterns
- Observing stock depletion rates
- Tracking supplier delivery consistency
- Identifying peak and low-demand periods

At the end of the analysis, Trend analysis allowed the prediction of high-demand products, planning of restocking cycles, and identification of months with low or high cash flow.

3.4.3 Pareto Analysis

Pareto Analysis is a statistical decision-making technique derived from the Pareto Principle, commonly known as the 80/20 rule. This principle

asserts that roughly 80% of outcomes result from 20% of causes, highlighting the unequal distribution of effort and impact in various systems (Koch, 1998). In practical business applications, Pareto Analysis helps organizations identify the vital few contributors, such as top-performing customers, products, or processes that generate the majority of results.

The origins of the principle trace back to Vilfredo Pareto, who, in the early 20th century, observed that approximately 80% of Italy's land was owned by 20% of the population (Pareto, 1896). This observation was later generalized to multiple domains, forming the basis for modern applications in business management, quality control, and strategic planning. In customer relationship management, for example, Pareto Analysis enables companies to identify the top 20% of customers responsible for 80% of sales revenue, guiding targeted marketing strategies and improving overall profitability.

Figure 2: Pareto Chart showing customer contribution to sales

In this study from Figure 2, Pareto Analysis was specifically applied to:

- Identify high-value customers generating the majority of sales revenue
- Determine the proportion of customers accounting for 80% of total sales

- **Segment customers based on their revenue contribution**

The Pareto Analysis was conducted through the following systematic steps:

Step 1: Data Compilation: **Customer sales data was extracted from the organization's database and aggregated to determine each customer's total revenue contribution during the analysis period.**

Step 2: Ranking: **Customers were ranked in descending order based on their sales contribution, from highest to lowest revenue generators.**



	A	B	C
1	TOP CUSTOMERS		
2	CUSTOMER	PERCENTAGE CONTRIBUTION	TOTAL SALES
3	WEST AFRICAN CERAMICS	24.83%	₦ 29,276,504.13
4	GRAND CEREALS LTD	9.65%	₦ 11,378,600.50
5	BUA PASTA AND FLOUR, PH	7.57%	₦ 8,926,184.66
6	DANGOTE PETROLEUM	7.55%	₦ 8,899,017.13
7	DUFIL PRIMA FOODS LTD	6.74%	₦ 7,950,944.25
8	EFE GLOBAL SERVICES	6.02%	₦ 7,093,460.80
9	(OLAM)OK FOODS	5.98%	₦ 7,049,750.35
10	HUPSTACK	5.75%	₦ 6,785,665.82
11	BUA SUGAR REFINERY LTD	4.26%	₦ 5,024,798.32
12	STERLING OIL EXPLORATION	2.95%	₦ 3,481,187.84
13	GREENPEG LTD	2.84%	₦ 3,344,000.00
14	EVERATI NIGERIA LIMITED	2.58%	₦ 3,042,610.93
15	MR OBARO	1.89%	₦ 2,231,214.89
16	MR KELVIN	1.55%	₦ 1,832,055.16
17	CHITEX	1.53%	₦ 1,809,336.45
18	SUNFLAG(NIGERIA) LIMITED	1.51%	₦ 1,785,218.47
19	STANDARD(PURE) FLOUR, A	0.91%	₦ 1,072,245.85
20	RAFFLES OIL LFTZ ENTERPRI	0.77%	₦ 908,744.23
21	AKACHI IND LTD	0.73%	₦ 860,207.56
22	AAVA BRANDS MANUFACTU	0.58%	₦ 680,569.06
23	MR JOHN	0.48%	₦ 567,422.63
24	FOLA'S TECHNICAL	0.41%	₦ 482,779.17
25	VASCO INDUSTRIES LTD	0.36%	₦ 421,655.07
26	MR DANIEL	0.31%	₦ 363,631.42
27	GEO-LIND NIG LTD	0.28%	₦ 327,660.70
28	LOMBARDY PLASTICS	0.27%	₦ 322,773.84
29	EMOBELLA ENGINEERING	0.27%	₦ 312,836.61
30	AMIDADA'S GLOBAL VENTUI	0.26%	₦ 301,301.00
31	MR. GODWIN	0.25%	₦ 300,000.00
32	ROSHAF ENGINEERING ENTE	0.22%	₦ 263,670.00
33	AKPOSAZ	0.20%	₦ 237,963.87
34	ZAIKA FOODS LTD	0.12%	₦ 137,604.57
35	BME GROUPE, LOME	0.09%	₦ 106,231.12

SALES | 2024 | P.O TRACK SYSTEM | EDA

Ready | Accessibility: Investigate

Figure 3: Ranking of customers based on their sales contribution

Step 3: Cumulative Calculation

For each customer, the following metrics were calculated:

- **Individual contribution percentage: (Customer Sales / Total Sales) × 100**
- **Cumulative revenue percentage: Running total of contributions from top to bottom**

Step 4: Customer Segmentation

Based on the analysis, customers were classified into three categories:

Category	Description	Contribution Range
		Top customers

Category A	**High-value customers**	**contribute to 80% of sales**
Category B	**Medium-value customers**	**Customers contributing to the next 15% of sales**
Category C	**Low-value customers**	**Remaining customers contributing to the final 5%**

Figure 4:Table of customer segmentation from Pareto Analysis

3.5 Systems Developed

During my Industrial Training at Elrol Integrated Services Ltd, I developed a series of integrated business systems designed to automate key operations and enhance efficiency within the Procurement and Inventory Department. These systems include the Sales Record System, Purchase Order (PO) Tracking System, Inventory Record System, and Incoming Stock Record System. Together, they function as a lightweight Enterprise Resource Planning (ERP) solution, interconnected through Microsoft Excel and synchronized via Microsoft OneDrive for real-time data updates and collaboration across departments.

The Sales Record System was developed to track daily sales transactions, capturing essential parameters such as *product name, quantity sold, sales value, discounts, and VAT payments*. Automated formulas and data validation were incorporated to ensure the accuracy of entries and eliminate redundancies. The system also includes summary tables and performance indicators that provide quick insights into sales trends and product performance.

The Purchase Order (PO) Tracking System was created to monitor procurement activities, supplier performance, and delivery timelines. It allows staff to record every purchase order from initiation to completion — tracking order status, supplier details, product quantities, and payment verification. By linking this record to the inventory sheet, updates occur automatically when goods are received or delivered.

The Inventory Record System maintains a real-time balance of all stock items in the warehouse. It combines data from sales, procurement, and incoming stock sheets to automatically update stock quantities. Conditional formatting highlights low-stock levels, while IF functions ensure that every item is properly referenced across all records. This eliminated the need for manual counting and reduced the risk of discrepancies between physical and recorded stock.

The Incoming Stock Record System serves as the entry point for tracking all goods received into the warehouse. It records supplier name, delivery date, quantity received, and product specifications. When new items are entered, the data automatically syncs with the inventory system to reflect the updated stock count.

All four systems are integrated using Excel formulas such as IF and IFS functions, as well as data validation and conditional logic. The systems were hosted on OneDrive, allowing multiple users to access, update, and share real-time information simultaneously.

To complement these systems, a Power BI dashboard was connected to the Excel datasets, providing visual analytics and performance indicators such as total sales, supplier performance, and product movement trends. This gave management an instant overview of company operations, enabling faster and more informed decision-making.

The introduction of this integrated system improved operational efficiency, reduced manual workload, and ensured data consistency across departments. It also provided a framework that can easily be expanded or migrated into a full ERP solution in the future as the company grows.

In addition to the core operational systems, I developed an AI-powered Customer Support System for Elrol Integrated Services Ltd. The system was trained with the company's product and service information, frequently asked questions, and operational procedures to provide instant responses to customer inquiries. This chatbot reduces response time, enhances customer engagement, and ensures that clients can access assistance 24/7 even in the absence of support staff. It can also be integrated into the company's website or social media platforms to automate communication and improve overall service delivery efficiency.

3.6 Tools and Software Used

During the course of my training, I utilized several tools and software to perform data analysis, automate processes, and build business systems:

- **Microsoft Excel (2019 & Online):** Used for data entry, cleaning, analysis, and building automated record systems (Sales, Inventory, Purchase Orders, etc.).
- **Power BI:** Used for creating live interactive dashboards connected to OneDrive for real-time business performance monitoring.
- **Microsoft OneDrive:** Served as a cloud database and central storage system that enabled multi-user access and synchronization across departments.
- **Chatbase (AI Chatbot Builder):** Used to develop an AI-powered customer support system that automates customer inquiries.
- **Power Query:** For data transformation and automation of updates between Excel and Power BI.

CHAPTER FOUR:

DATA ANALYSIS

4.1 Introduction

This chapter presents the analysis of operational data at Elrol Integrated Services Ltd. The analysis was carried out to understand the performance of the company in terms of sales and procurement management. The primary objective was to generate actionable insights to guide business decisions, improve operational efficiency, and support management in strategic planning.

4.2 Data Preparation and Cleaning

Before analysis, the datasets were cleaned and prepared to ensure accuracy:

I. Duplicate entries and blank fields were removed.

II. Date formats and numeric values were standardized for consistency.

III. Product and customer names were normalized to avoid mismatches.

IV. Totals and quantities were cross-validated between POs, inventory, and sales records.

This preparation ensured reliable descriptive and trend analyses.

4.5 Exploratory Data Analysis

Descriptive Statistics provided a clear overview of the dataset through measures such as totals, percentages, and averages. This allows management to quickly identify high-performing products, customers, and periods.

Trend Analysis involves plotting data over time to identify patterns, peaks, and declines. In this context, it helped track monthly sales fluctuations, product movement trends, and recurring customer activity.

- **Top Selling Products**

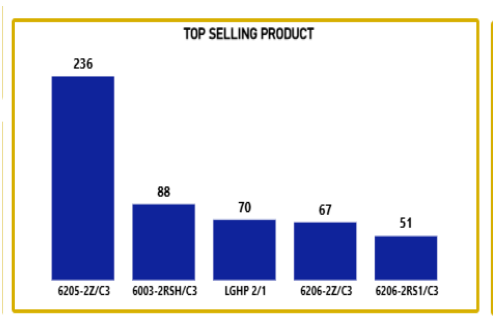


Figure 5:Top selling products

The analysis in Figure 5 identified the products generating the highest amount of market demand. These insights helped prioritize stock planning, procurement, and marketing efforts to focus on highly demanded items.

- **Top Customers**

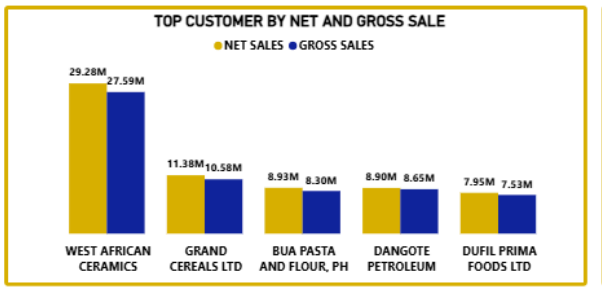


Figure 6: Top customers by Net and gross sales

From Figure 6, Customers were ranked based on net sales and gross sales. Identifying high-value customers allowed for strategies like contract supply agreements and loyalty incentives to secure repeat business and minimize risk if any major customer reduces orders.

- **Most Frequent Customers**

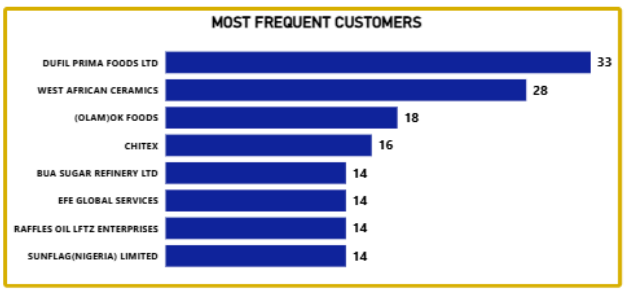


Figure 7: Most Frequent customers

Frequent buyers were tracked to understand loyalty and purchasing behavior. Recommendations included strategies to increase average order value per transaction, such as bundle offers, volume discounts, and targeted promotions.

4.6 Interpretation of Trend Analysis

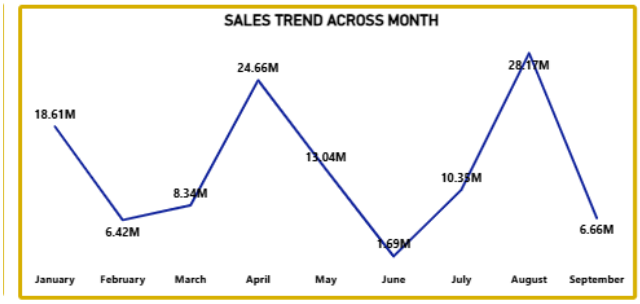


Figure 8: Sales trend across months

Monthly sales trends were analyzed in Figure 8 to identify irregularities, peaks, and dips. Although 2025 was the first available dataset, the trends provided insights for inventory management and procurement planning.

The line charts revealed irregular sales trends across months, highlighting periods of high and low sales. This trend analysis indicates which months contribute most to revenue and helps management make informed decisions about marketing strategies, and stock management. Although the dataset only covers a 9 months, the visualized trends provide actionable insights into business performance and inform recommendations for future growth strategies.

- **Key Performance Indicators (KPIs)**



Figure 9: Key Performance Indicators

KPIs such as total sales — Net & Gross, quantity sold, and Taxes balance were visualized in the Power BI dashboard. The dashboard enabled real-time monitoring for the MD and management team, supporting immediate, data-driven decisions.

4.7 Interpretation of Results

I. Sales are concentrated among a few key customers, which poses a risk if a major customer reduces their orders.

II. Certain high-value, high-demand products require targeted marketing to grow the customer base and increase revenue.

III. Frequent customers could benefit from loyalty programs and strategies that encourage larger orders.

IV. Real-time dashboards enable management to respond quickly to stock shortages, procurement delays, and sales opportunities.

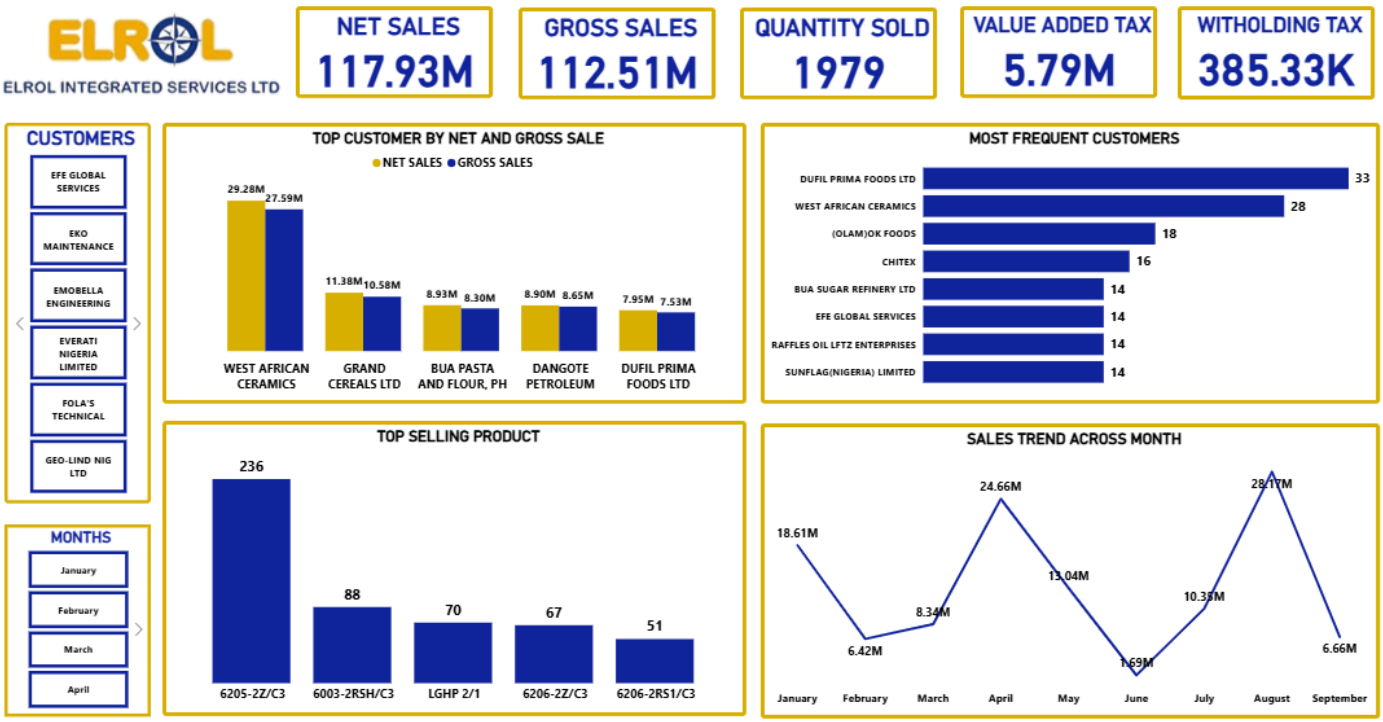


Figure 7: Powe-BI dashboard of the analyses

CHAPTER FIVE:

CONCLUSION & RECOMMENDATION

5.1 Summary of Experiences:

During my Industrial Training at Elrol Integrated Services Ltd, I was mainly assigned to the Procurement and IT departments, where I applied my skills in data analytics and business system development. My primary duties included analyzing sales data, creating automated business systems, and providing actionable insights to management. I conducted a detailed analysis of the company's sales data, focusing on top-selling products, key customers, most frequent buyers, product movement in the market, and sales trends over time. This was done using descriptive statistics and trend analysis, with the findings displayed on an interactive Power BI dashboard presented to the Managing Director. Additionally, I developed several automated systems, including:

• Sales Record System – centralizes sales data for easier tracking and analysis.

• Purchase Order (PO) Tracking System – tracks orders from start to finish.

• Inventory Record and Incoming Stock System – monitors stock levels and movement with real-time updates.

These systems were connected through Microsoft OneDrive, creating a simplified, ERP-like centralized solution for the company. Automating stock-out processes, linking POs and inventory records, and developing a live management dashboard greatly enhanced operational efficiency.

5.2 Skills Acquired:

During the SIWES program, I developed and strengthened the following skills:

- **Data Analytics:** Using Excel and Power BI for data cleaning, visualization, KPI monitoring, and sales trend analysis.
- **Business System Development:** Designing and implementing interconnected automated systems for sales, purchase orders, and inventory.
- **Problem-Solving & Decision Support:** Identifying business gaps, interpreting data, and providing actionable recommendations to improve efficiency and revenue.
- **Professional Communication:** Presenting complex analyses and insights to management

in an understandable and actionable manner.

5.3 Conclusion:

The training program provided me with a practical understanding of how data analytics can influence business decisions and improve operational efficiency. By developing automated systems and providing management with actionable insights, I was able to demonstrate tangible value and contribute meaningfully to Elrol Integrated Services Ltd. The experience also reinforced the importance of structured data collection, real-time monitoring, and effective reporting in a business environment.

5.4 Recommendations:

Based on my observations and analyses during the training period, the following recommendations are proposed for the company:

1. **Expand Customer Base:** Leverage digital marketing and social media to acquire new clients, increasing sales revenue and reducing dependence on a few top customers.
2. **Implement Loyalty Programs:** Engage frequent customers through consistent communication, bundle offers, volume discounts, and loyalty incentives to increase order value per transaction.
3. **Monitor Inventory Closely:** Ensure incoming stock is promptly recorded and integrated into the inventory system for accurate stock tracking and decision-making.
4. **Regular System Updates:** Continuously maintain and update the automated systems to reflect business changes, including adding new products, suppliers, or departments.
5. **Consider AI Integration:** Expand the use of AI tools, such as the customer support chatbot, to enhance customer experience, reduce response time, and improve operational efficiency.

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